

★ WEEK 1 · PART 3 of 3

Random Walks & Market Efficiency

Is the Past Any Guide to the Future?

Random walk · martingale · the Efficient Market Hypothesis · anomalies

Part 1 gave us returns; Part 2 gave us their patterns. Part 3 asks the deepest question: how much of the future can we actually predict — and why the market resists us.

3

PART 3 · EFFICIENCY

Why Markets Resist Prediction

From the random walk to the Efficient Market Hypothesis — the theory of why beating the market is so hard, and the evidence for and against it.

The Mathematician Who Beat the Market

Edward O. Thorp — a story that frames the whole debate

A math professor first beat casino blackjack with card counting, then turned the same rigor on Wall Street — and quietly outperformed for nearly two decades.

The setup

- Thorp proved blackjack was beatable — casinos changed their rules.
- He then asked: are financial markets beatable too?
- His tool: quantify mispricing in warrants and options, and hedge it.

The claim to test

- The reigning academic view: prices follow a random walk.
- If true, no strategy should beat the market on a risk-adjusted basis.
- So who is right — the theory, or the professor?

The Resolution, and the Lesson

What Thorp's record actually teaches us

His fund Princeton/Newport Partners ran 1969–1988, using hedged warrant and option arbitrage to earn strong, remarkably steady returns with no losing year.

So markets can be beaten?

- Yes — but only through narrow, identifiable, hedged edges.
- Not by predicting direction; by pricing risk better than others.
- And such edges shrink as more people find them.

The lesson for us

- Efficiency is a powerful baseline, not an absolute law.
- The bar for 'an edge' is very high — most apparent edges are noise.
- This tension runs through the entire course.

The Central Question of Part 3

Where, exactly, does prediction become possible?

1

If prices already reflect all information, the past is useless

That is the core intuition of market efficiency: whatever is knowable is already in the price, so the next move is essentially unpredictable.

2

The random walk and the martingale make this precise

These two models formalize 'unpredictable.' We will see exactly what each one assumes — and how they differ.

3

Reality is messier than the theory

Anomalies, bubbles, and Thorp's own record show the theory is an approximation. The debate is about how good an approximation.

Efficiency is the null hypothesis; the whole field is a series of attempts to reject it.

The Random Walk Hypothesis

Tomorrow's price = today's price + drift + an unpredictable shock

$$P_t = \mu + P_{t-1} + \varepsilon_t, \quad \varepsilon_t \text{ i.i.d.}, \quad \mathbb{E}[\varepsilon_t] = 0$$

What it says

- The best forecast of tomorrow's price is today's price (plus drift).
- Shocks are unpredictable with zero mean — no exploitable pattern.
- In returns: $r_t = \text{drift} + \text{noise}$, with no memory.

Why it matters

- It is the simplest model of an unpredictable price.
- It implies technical analysis of past prices cannot add value.
- It is the benchmark every 'edge' must beat.

What a Random Walk Looks Like

Nine simulated paths from an identical starting point. Each looks like it has 'trends' — yet none was foreseeable. Our eyes invent patterns that were never there.

Nine random walks from the same start — each equally plausible, none foreseeable



Three Flavors of Random Walk

'Unpredictable' comes in strengths — and only the weakest fits real data

Version	Shocks (increments) are ...	What it rules out
RW1 (strongest)	independent AND identically distributed	any dependence whatsoever
RW2	independent, but not identically distributed	any dependence; allows a changing mean/variance
RW3 (weakest)	merely uncorrelated (may still be dependent)	only LINEAR dependence

Key link to Part 2: real returns fit only RW3 — uncorrelated, yet dependent through volatility clustering.

The Martingale Property

A precise statement of 'a fair game'

$$\mathbb{E}[P_{t+1} \mid \mathcal{F}_t] = P_t \quad (\text{martingale: a fair game})$$

Reading the notation

- $\mathcal{F}_t$ is the information available today.
- Given everything known now, the expected future price is today's.
- So no strategy using today's info earns an expected edge.

Fair game intuition

- A martingale is the math of a bet with no expected gain.
- Excess returns form a 'martingale difference': mean zero given the past.
- This is the cleanest formal version of market efficiency.

Martingale vs Random Walk

A subtle but important difference

Random walk (stronger)

- Restricts the WHOLE distribution of shocks (independence).
- Rules out any dependence, including in the variance.
- Too strong: real returns violate it via volatility clustering.

Martingale (weaker)

- Restricts only the CONDITIONAL MEAN (expected next move = 0).
- Says nothing about variance — clustering is allowed.
- This is why efficiency is usually stated as a martingale, not a random walk.

From Random Walk to Market Efficiency

The economic idea behind the mathematics

Prices are unpredictable not by accident, but because competition among informed traders forces all knowable information into the price almost instantly.

The mechanism

- Any predictable profit attracts traders who trade it away.
- Their trading moves the price until the profit disappears.
- What remains is an unpredictable, 'fair-game' residual.

The consequence

- Prices reflect information — that is the Efficient Market Hypothesis.
- The random walk is the statistical shadow of this economic force.
- Now we make 'reflect information' precise.

The Efficient Market Hypothesis (EMH)

Prices fully reflect the available information set

$$\mathbb{E}[r_{t+1} - \mathbb{E}(r_{t+1} | I_t) | I_t] = 0 \quad (\text{prices reflect information})$$

In words

- You cannot systematically beat the market using information in I_t .
- Whatever is in I_t is already priced; only surprises move prices.
- By definition, surprises are unpredictable.

The key subtlety

- Efficiency is always relative to an information set I_t .
- Different information sets give different, testable forms (next).
- 'Efficient' never means 'the price is correct' — only 'unbeatable'.

Three Forms of Market Efficiency

Each form widens the information set already priced in. The wider the set, the harder the market is to beat.

Three Forms of Market Efficiency

Each form widens the information set already priced in. Bigger set = harder to beat the market.



What each form says you CANNOT do:

- **Weak:** beat the market using charts of past prices.
- **Semi-strong:** profit from public news — it is already in the price by the time you trade.
- **Strong:** profit even on inside information (rejected in practice — insiders do earn abnormal returns).

Most evidence supports weak & semi-strong; strong form is genera

Testing Weak-Form Efficiency

Can past prices predict future returns? Three classic tests

$$VR(q) = \frac{\text{Var}(r_t(q))}{q \cdot \text{Var}(r_t)} = 1 \quad \text{under a random walk}$$

The tests

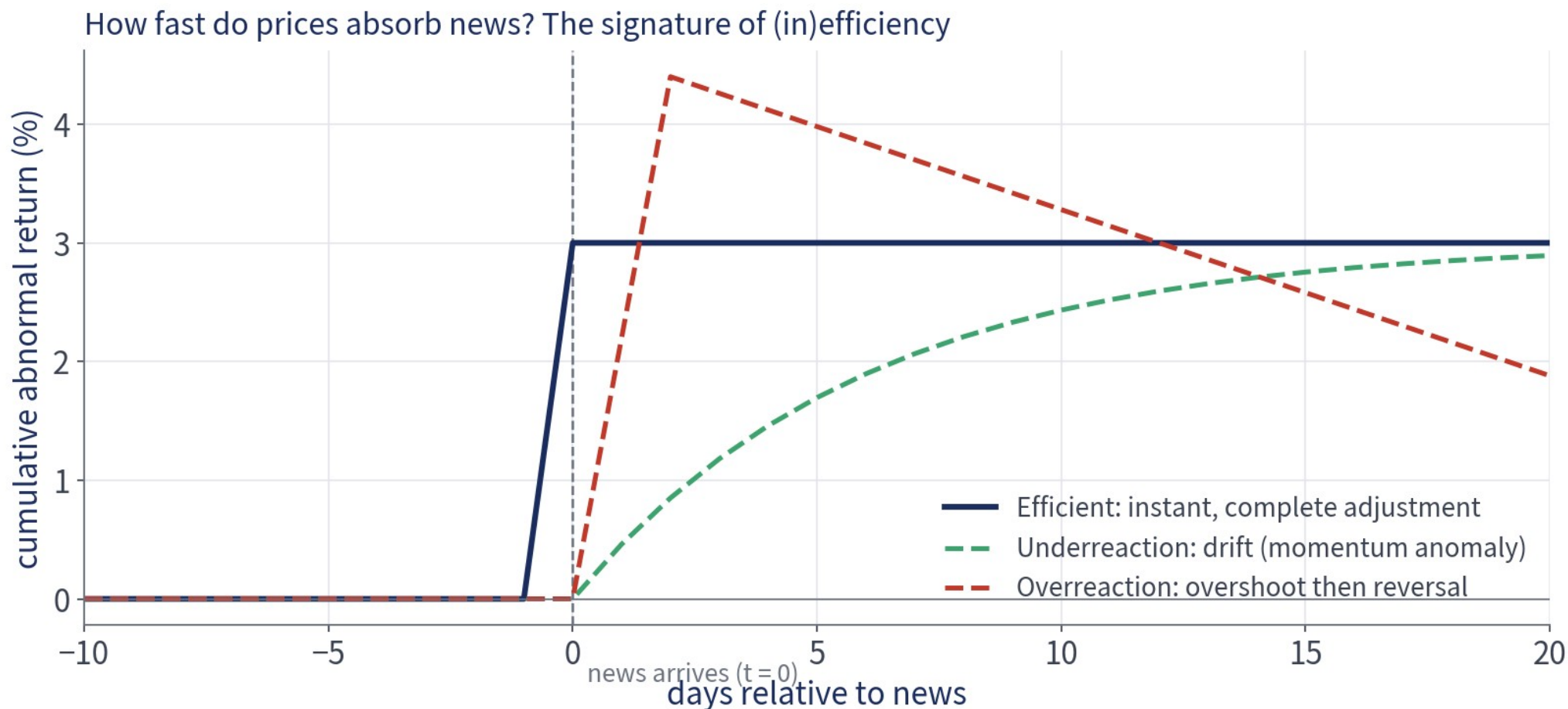
- Autocorrelation: is $\rho_R(k)$ different from zero? (Part 2: no.)
- Variance ratio: does variance scale linearly with horizon?
- Runs test: are up/down streaks more ordered than chance?

The verdict

- Under a random walk, the variance ratio equals 1 at all horizons.
- Broadly, weak-form efficiency holds well for liquid markets.
- Deviations exist but are small and often unstable.

Testing Semi-Strong Form: How Fast Do Prices React?

An efficient market jumps once, instantly and completely (navy). Underreaction (drift) and overreaction (reversal) are the footprints of inefficiency.



The Joint Hypothesis Problem

Why efficiency can never be tested on its own

$$\underbrace{\text{test of efficiency}}_{\text{what we want}} = \underbrace{\text{efficiency}}_{\text{joint hypothesis}} \text{ and } \underbrace{\text{a model of } \mathbb{E}[r]}_{\text{joint hypothesis}}$$

Fama's insight

- To test efficiency you need a model of 'normal' expected returns.
- So every test is really a test of TWO things at once.
- A rejection could mean inefficiency — or a wrong return model.

Why it matters

- 'Anomalies' may just be a mispriced risk factor we haven't modeled.
- This makes the efficiency debate genuinely hard to settle.
- It keeps both camps honest — and the question open.

Evidence That Markets Are Efficient

Why the baseline is so hard to beat

What we observe

- Most active managers underperform simple index funds after fees.
- Persistent, repeatable alpha is rare — winners rarely stay winners.
- Prices react to public news within seconds, not days.

Why it holds

- Millions of competing, informed traders leave little on the table.
- Low-cost information and fast execution close gaps quickly.
- The bar Thorp had to clear shows just how efficient markets are.

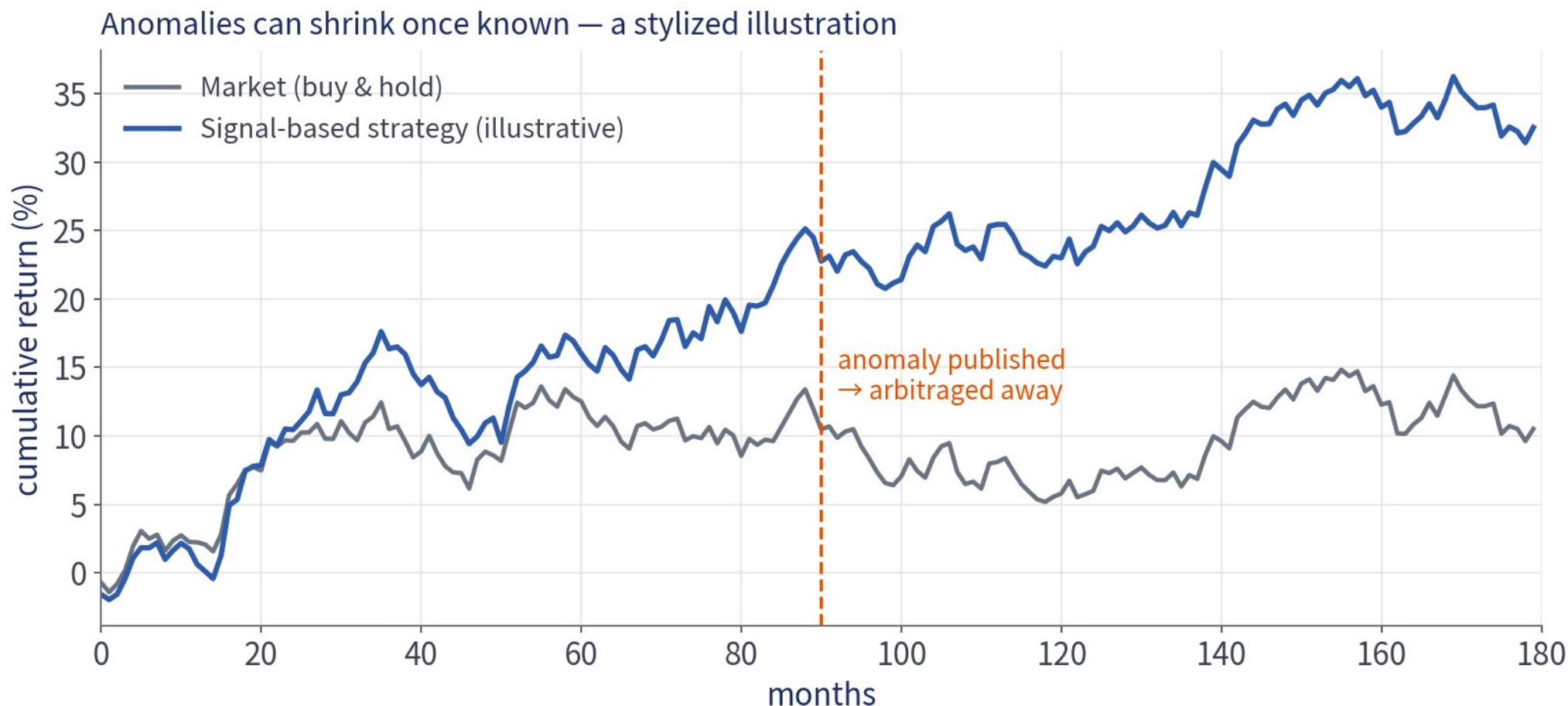
The Anomalies

Patterns the strict theory struggles to explain

Anomaly	The observed pattern	Efficient-market reply
Momentum	Past winners keep winning (3–12 months)	A risk premium — or it fades once known
Value	Cheap (low P/B) stocks tend to outperform	Compensation for distress risk
Size	Small-cap stocks historically outperform	Weak and unstable after publication
Calendar effects	Returns cluster in certain periods	Often vanish out of sample
Post-earnings drift	Prices drift after earnings surprises	Underreaction plus limits to arbitrage

Anomalies Often Fade Once They Are Known

A stylized illustration: a profitable signal loses its edge after publication, as arbitrageurs pile in — exactly what an efficient market predicts.



The Behavioral Challenge

Where human psychology bends prices

Systematic biases

- Overreaction: crowds chase, then reverse (bubbles and busts).
- Underreaction: news is absorbed slowly, creating drift.
- Herding, anchoring, and loss aversion push prices off value.

Why mispricing can persist

- Limits to arbitrage: correcting a mispricing is costly and risky.
- 'Markets can stay irrational longer than you can stay solvent.'
- So biases are not always competed away immediately.

The Grossman–Stiglitz Paradox

A market cannot be perfectly efficient

If prices already reflected all information, no one would be paid to gather it — but then who puts the information into prices? Perfect efficiency is self-defeating.

The logic

- Gathering information is costly.
- In a perfectly efficient market, that effort earns nothing.
- So rational traders would stop gathering — and efficiency collapses.

The resolution

- Markets are 'efficiently inefficient': just enough profit to pay analysts.
- A small, hard-won edge is the reward for making prices informative.
- This is precisely the space Thorp operated in.

The Adaptive Markets Hypothesis

Efficiency as an evolving, not fixed, property

The idea (Lo)

- Efficiency is not binary; it waxes and wanes with conditions.
- Investors adapt, strategies compete, and edges are born and die.
- It blends efficient-market logic with behavioral realism.

What it implies

- Anomalies can appear, be exploited, and disappear over time.
- No strategy works forever; adaptation is the only constant.
- This reconciles Thorp's edge with the market's overall efficiency.

Efficiency vs Reality

The debate that never fully closes

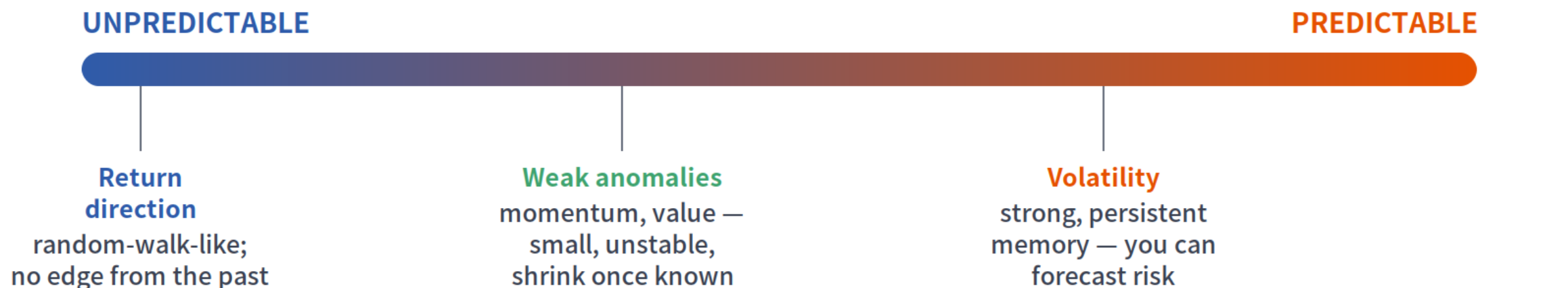
Question	Efficient-market view	Behavioral / reality view
Can you beat the market?	No — prices are fair, net of costs	Sometimes, via narrow identifiable edges
Why do anomalies exist?	Risk premia, or data mining	Real mispricing from human bias
Are investors rational?	On average and in aggregate, yes	Systematically biased at times
Do prices always reflect value?	Yes, and quickly	Bubbles and crashes suggest not

The Predictability Spectrum

Efficiency is not all-or-nothing. Return direction sits near the unpredictable end; volatility sits near the predictable end — the throughline of the whole week.

The Predictability Spectrum

Efficiency is not all-or-nothing. Different quantities sit at different points on this line.



Takeaway: the DIRECTION of returns is near the unpredictable end; the SIZE of returns (volatility) is near the predictable end.

Putting Efficiency to the Test

Direct Claude Code to run the weak-form efficiency tests on real data

GOAL Test weak-form efficiency on your own returns — and see the joint-hypothesis caveat in action.

[INPUT] Prompt · Claude Code

Using returns.csv, write and run efficiency.py:

1. autocorrelation of returns at lags 1..20 with 95% confidence bands;
 2. the variance ratio $VR(q)$ for $q = 2, 5, 10, 20$, testing whether $VR(q) = 1$;
 3. a runs test on the sign of daily returns.
- For each test, print the statistic and a plain-English verdict on weak-form efficiency, and state the joint-hypothesis caveat. Save the ACF and variance-ratio plots to /figs.

How to read it

- $VR \approx 1$ and $\rho \approx 0$ support weak-form efficiency.
- A rejection may mean a wrong return model.
- Try an illiquid ticker — do results change?
- Evidence, never proof — stay skeptical.

Deliverable: efficiency.py + a short note on whether your asset looks weak-form efficient.

Three Questions Before You Move On

If you can answer these, Part 3 has landed

Q1

Real returns are uncorrelated yet dependent. Which random-walk version fits?

RW3 (weakest)

Q2

Why can market efficiency never be tested by itself?

Joint hypothesis

Q3

Efficiency says the past is useless. Does that contradict volatility forecasting?

No — size $\neq$ sign

All three tie Part 3 back to the stylized facts of Part 2.

Week 1 in Three Ideas

Pulling the whole week together

1

Speak in returns — and prefer log returns

Part 1: prices are non-stationary; returns are the tractable unit. Match the mean to the question, and respect the traps.

2

Real returns have a fingerprint

Part 2: fat tails, negative skew, and clustering volatility. Direction is unpredictable; magnitude is not.

3

Markets are efficient — but not perfectly

Part 3: the random walk and EMH explain why prediction is hard, while anomalies and Thorp show the theory is an approximation.

Next — Week 2: Stationarity and Unit Roots, making 'roughly stationary' precise.